Anurag Bheemani

Machine Learning Engineer | Computer Vision & Applied GenAI

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Summary

Machine Learning Engineer focused on computer vision and video understanding. Hands-on with self-supervised video models (V-JEPA), object detection (YOLO), and model deployment via TensorRT and DeepStream, on a solid Python and backend foundation.

Experience

Eizen AI - Machine Learning Engineer — Nov 2025 - Jun 2026
Computer Vision, Hyderabad

- Trained a lightweight attentive probe on Meta's frozen V-JEPA 2.1 video encoder for 5-class bear-behavior recognition: 75.5% supervised probe accuracy and 72.6% weighted F1, with 66.9% purity from unsupervised clustering of the same embeddings; deployed the model to NVIDIA Jetson edge devices for on-site inference.
- Benchmarked a DeepStream 9.0 with TensorRT (FP16) pipeline against a PyTorch baseline for a 375.5M-parameter V-JEPA 2 encoder (PyTorch to ONNX to TensorRT, NVDEC decode, 5D batching): 28.9 ms vs 211.9 ms inference latency per 16-frame clip (7.2x faster) and 39% less GPU memory; verified ONNX-export compatibility across video models (r3d_18, VideoMAE, V-JEPA 2) to guide the team's implementation.
- Developed a kitchen procurement-verification POC: YOLOv8-OBB seven-segment digit recognition (precision 0.95, recall 0.95, mAP@50 0.96 on standard data; 81% in the field due to domain shift from printed-digit training data) with one-shot Hailo CLIP item recognition on a Raspberry Pi; used the Gemini API to extract expected item weights from procurement sheets and match them against live scale readings to flag discrepancies.
- Prototyped a video-to-motion behavior-cloning POC in NVIDIA Isaac Sim (AMPLIFY) for an industrial robot arm's pick-and-place: CoTracker point tracking, a motion tokenizer, forward-dynamics action prediction, UniDepth depth estimation, and inverse kinematics on the robot URDF.
- Improved a leopard-detection model's accuracy from 83% to 92% by expanding its dataset through DINO-assisted auto-annotation with human validation sampling, and deployed the model to NVIDIA Jetson edge devices for on-site inference.
- Automated dataset labeling with a cron-triggered pipeline that checks edge-device local storage every 10 minutes and dispatches image batches (about 2,000) to a DINOv2 and SAM auto-annotation workflow.
- Engineered a semi-automatic annotation tool combining model-assisted pre-labeling with manual correction, with multi-user project allocation, batched assignment, and zip export.
- Migrated OCR from Tesseract to PaddleOCR for layout-aware UI extraction, achieving 91% word accuracy across 150+ cross-platform system-monitoring screenshots.

GLOOMDEV - Web Developer | *Intern* — July 2024 - Dec 2024
Full Stack Web Developer (Backend Emphasis)

- Developed gloom-dev.com within the first month, focusing on backend integration and deployment workflows.
- Delivered a scalable full-stack e-commerce platform using the MERN stack with secure API and database architecture.
- Integrated the Razorpay payment gateway with backend endpoints, completing 50+ test transactions in staging.
- Built a basic conversational chatbot using LangChain with the Gemini API as the backend LLM.

Skills

Languages: Python, C++, JavaScript, SQL, TypeScript
ML & Deep Learning: PyTorch, scikit-learn, NumPy, pandas, Hugging Face Transformers, Vertex AI
Computer Vision: YOLOv8 (OBB), OpenCV, MediaPipe, SAM, DINOv2, CLIP, PaddleOCR, Whisper, V-JEPA
LLM & GenAI Tools: Gemini SDK, LangChain, ChromaDB, RAG, llama.cpp (GGUF), whisper.cpp (GGML)
Inference & Deployment: TensorRT, NVIDIA Triton, NVIDIA DeepStream SDK, ONNX / ONNX Runtime, NVDEC, Docker, CUDA, Vulkan
Simulation & Edge: NVIDIA Isaac Sim, Nvidida Jetson Nano, Raspberry Pi (Hailo accelerator)
Backend & Data: FastAPI, Tauri, Node.js, REST APIs, Kafka, MongoDB, MySQL, PostgreSQL, Redis, GCP Cloud Run
Tools: Git, Linux

Projects

Orieon - Offline AI Video Player | Link — *Tauri, React, C++, llama.cpp, whisper.cpp, FastAPI*

- Built a commercial offline AI desktop video player for Windows using Tauri, React, C++, and Rust, supporting automatic transcription, translation (14+ languages), semantic search, AI chat, and document analysis.
- Integrated whisper.cpp and llama.cpp with CUDA, Vulkan, and CPU backends, delivering up to 5-10x faster local AI inference than CPU-only execution.
- Engineered a fully offline C++ RAG pipeline with custom ONNX embeddings, binary vector indexing, semantic transcript retrieval, query-intent classification, and persistent model serving to eliminate per-request loading overhead.
- Developed secure model distribution with encrypted GGUF/ONNX assets, in-memory decryption, HMAC-signed licensing, multi-factor hardware fingerprinting, and cross-language license validation (JavaScript, Rust, C++).

- Built the complete backend using FastAPI, PostgreSQL, and Google Cloud Run, implementing authentication, licensing, model delivery, payment processing (Razorpay & Paddle), and signature-verified webhooks.

- Optimized model packaging, downloading, and media processing for production deployment, delivering a fully functional Windows installer with automatic model management.

Hand Sign Recognition | Link *Python, MediaPipe, OpenCV, scikit-learn*

- Implemented a real-time hand-gesture recognition system: MediaPipe extracts 21 hand landmarks per frame, normalized into feature vectors and classified by a Random Forest, with no GPU or special hardware.
- Ran live on a standard webcam at about 86% accuracy with low, interactive latency.

Face Sorter - Photo Organizer | Link *Python, FastAPI, React Native, MongoDB*

- Developed a React Native + FastAPI app that groups photos by face using the `face_recognition` library (dlib 128-D encodings, Euclidean-distance matching), stored in MongoDB.
- Grouped 100+ photos per batch at about 70% correct grouping, removing manual sorting.

Education

Jawaharlal Nehru Technological University Hyderabad Dec 2021 - Jul 2025
B.Tech, Computer Science (AI & Machine Learning) *CGPA: 6.98/10.00*

Achievements

- "Exceptional Intern of the Month" during web-development internship at GLOOMDEV.
- Led 24 members across 6 groups for the technical execution of the college annual tech fest.

Certification

- Web Development Internship – GLOOM DEV
- Edunet Code Unnati Advance Course – EDUNET
- Code Unnati Innovation Marathon 2025 – EDUNET